

Henry Lee

Lead Software Engineer · Backend Platforms & Distributed Systems

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8 years SOFTWARE ENGINEERING	50k → 400k+ DAILY ACTIVE USERS	10M+ / day DATA RECORDS PROCESSED	100+ SDK RELEASES
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SUMMARY

Lead Software Engineer with 8 years of experience building and scaling backend systems. Progressed from production troubleshooting into owning core platform services and leading four engineers. Helped scale a high-concurrency transaction platform from **50k to 400k+ daily active** users, with recent work focused on backend platforms, reliable workflows, and developer tooling.

EXPERIENCE

SNSoft Sdn Bhd

- **Lead Software Engineer (Backend Team Lead)**

Dec 2025 — Present

 - Lead **four engineers** across six backend domains, owning solution design, work breakdown, and cross-team delivery while helping engineers independently lead their areas.
 - Led the rebuild of the platform's core transaction workflow using **NestJS, gRPC, and Temporal**, replacing a legacy synchronous flow with retry-safe processing for **millions of transactions daily**.
 - Consolidated transaction data from more than ten upstream sources into a unified pipeline processing over **10M records per day** across backend, data, and reporting teams.
 - Built and evolved the shared SDK and service contracts used by product teams, with **100+ releases** and onboarding reduced from **30-60 minutes to around 2 minutes**.
 - Improved production reliability through observability, graceful service handling, and database tuning, expanding analysis from one-day timeouts to reliable **30-day windows**.
 - Set integration boundaries and ownership across Frontend, BI, Payment, Provider, and AI teams, giving engineers clearer responsibility for design, delivery, and production support.
- **Senior Software Engineer**

Jan 2024 — Dec 2025

 - Built **LiveChat from the ground up** as the company's first production service on the **NestJS/gRPC** stack, starting solo and setting up the backend, Matrix messaging, data layer, and deployment.
 - Designed routing, queueing, concurrency safeguards, AI-to-human escalation, and operational tools; established reusable patterns and mentored **three junior engineers** to independent ownership.
 - Reworked critical bottlenecks as the platform grew from **50k to 400k+ daily active users**, moving real-time fan-out, counters, and hot-path state onto Redis through staged production changes.
 - Built large-volume reporting before a dedicated BI function existed, supporting operational analysis, compliance reporting, and the later reporting platform.

Software Engineer

Dec 2019 — Jan 2024

- Co-delivered platform-wide **multi-currency transaction support** across deposits, withdrawals, and transfers.
- Delivered regulatory reporting workflows for **million-row datasets**, including automated file splitting under strict submission deadlines.
- Integrated third-party providers and improved live-data reliability through WebSocket heartbeat, reconnection, and failure handling.
- Built operational reporting capabilities that later contributed to the platform's broader reporting foundation.

Hitachi Ebworx — Mobile Engineer

Apr 2018 — Oct 2019

- Built Great Eastern Singapore's first insurance **point-of-sale mobile application**, supporting agents in their daily sales workflow.
- Developed responsive web and hybrid mobile features across delivery, bug fixing, testing support, and production deployment.
- Owned QA deployments and **automated the daily release process** to improve speed and consistency.

EARLY CAREER

QA & Field-Test Engineer Intern, OPPO Electronics Malaysia HQ - mobile field testing, troubleshooting, test-case design, and product-quality feedback (2017).
IT & Network Support Intern, Chin Hin - IT and network support across 23 branches (2014).

SKILLS

LANGUAGES	TypeScript · JavaScript · SQL · Java (working) · Rust (working)
BACKEND	Node.js · NestJS · gRPC · Protocol Buffers · Express · WebSocket · Temporal · Apache
DATA	MongoDB (aggregation, indexing, Atlas Search) · PostgreSQL · Redis (Pub/Sub, Cluster) · ClickHouse · Hologres · Apache Flink · change data capture
OBSERVABILITY & INFRA	Grafana · Alibaba Cloud SLS · OpenTelemetry · Prometheus · HashiCorp Vault · Docker · GitHub Actions · Alibaba Cloud
QUALITY	Jest · Testcontainers · JUnit 5 · JaCoCo · integration testing · coverage gates · release regression
AI ENGINEERING	Claude Code · MCP · AI-assisted delivery workflows · context engineering · LLM application development
PRACTICES	contract-first APIs · event-driven architecture · idempotent processing · saga & compensation · domain-driven design · zero-downtime migrations

EDUCATION

Bachelor of Information Technology - HELP University · 2017 · English · Mandarin · Malay